

Electronic Supporting Material (ESM): Exploring groundwater and soil water storage changes across the CONUS at 12.5 km resolution by a Bayesian integration of GRACE data into W3RA

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Introduction

This file contains the following supporting information:

1. Details of MCMC-DA approach
2. Comparison between GRACE and W3RA TWSC within the Contiguous United States (CONUS)
3. Supplementary figures
4. Correlation coefficients between groundwater storage changes and the USGS groundwater level observation
5. Correlation coefficients and the Root Mean Squares of Differences (RMSD) between model-derived soil water storage changes and the ESA CCI soil moisture products
6. Independent Component Analysis (ICA) to extract the ENSO modes from water storage changes

ESM- Section 1. Method

A Bayesian Marcov Chain Monte Carlo-Data Assimilation (MCMC-DA) approach is formulated which is used to recursively update individual water storage components from the W3RA (as a priori information) with GRACE TWSC (as observations). This is done in a multi-variate analysis using a linear ‘state-space model’ (Bernstein, 2005), while both state parameters and error covariance matrix of the additive innovations are allowed to vary in time. In this formulation, the unknown state parameters and temporal dependency between them are estimated using the MCMC (Geyer, 1991; Gilks et al., 1996) algorithm, where the joint posterior distribution of the unknown parameters is estimated using the forward-filtering backward-smoothing recursion approach as in Kitagawa (1987). In what follows, details of the MCMC-DA algorithm and the differences with DMDA (Mehrnegar et al., 2020) are described, where the latter is previously used for merging an ensemble of water storage simulations from multi-model outputs and GRACE TWSC.

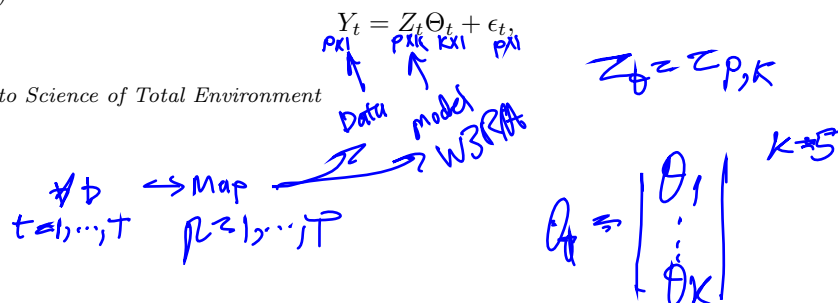
ESM- Section 1.1. Multi-variate linear state-space model

Linear relationships between GRACE TWSC and the modeled individual water storage components from W3RA can be represented by a multi-variate state-space model known as the ‘observation equation’ (Bernstein, 2005):

$$Y_t = Z_t \Theta_t + \epsilon_t, \quad (\text{ESM.1})$$

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① 3. Estimate for each point! how in relation to SWCN?
 1. Spatial Variability can be learned
 2. Clustering and Segmenting



Estimate: 5 state parameters for each map, $(\gamma_1, \dots, \gamma_P)$

! too sensitive to the spatial extent.

where $Y_{tP \times 1}$ represents the GRACE TWSC (observations) for P spatial grid points, at time $t = 1, 2, \dots, T$. In Eq. (ESM.1), the water storage states of W3RA are stored in a diagonal matrix $Z_{tP \times P}$, where the diagonal elements of Z_t contain $[z_{1,t}, z_{2,t}, \dots, z_{K,t}]$ corresponding to each spatial grid point $p = 1, 2, \dots, P$, and $K = 5$ is the number of individual water storage compartments simulated by W3RA, i.e., surface water storage, surface soil water storage (top layer), shallow rooted and deep rooted soil water storage (according to model description, Van Dijk, 2010), as well as groundwater storage. Each element of matrix $\Theta_{tP \times 1}$ contains the unknown "state parameters", $[\theta_{1,t}, \theta_{2,t}, \dots, \theta_{K,t}]^T$ corresponding to each spatial grid point $p = 1, 2, \dots, P$, which make a linear relation between GRACE TWSC (Y_t) and the model's individual water storage components (Z_t) in Eq. (ESM.1). In an ideal situation, where the GRACE TWSC is equal to the summation of individual water storage components of the model, the regression parameters Θ_t are equal to 1. However, it is expected that the model estimates contain errors due to the imperfect structure of the hydrological model and uncertainties in the input forcing data that are used to run the model simulations. Besides, GRACE TWSC contain errors as described, e.g., in Forootan et al. (2014). As a result, the state parameters Θ_t are unknown, and they are allowed to evolve in time according to the 'state equation':

$$\Theta_{t+1} = \Theta_t + \delta_t \quad \text{(ESM.2)}$$

In Eqs. (ESM.1) and (ESM.2), ϵ_t and δ_t are additive innovations (i.e., residual) that correspond to the observation equation (Eq. (ESM.1)) and the state equation (Eq. (ESM.2)), respectively. These residuals are assumed to be Gaussian distributed and independent from each other with a mean value of zero and an error covariance matrix of V_t and Q , respectively. Thus, the distribution can be written as:

$$\begin{bmatrix} \epsilon_t \\ \delta_t \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} V_t & 0 \\ 0 & Q \end{bmatrix} \right). \quad \text{(ESM.3)}$$

Uncertainty in GRACE TWSC is reflected in V_t and contains the spatial correlations as suggested by Schumacher (2016). The error covariance matrix Q in Eq. (ESM.2) corresponds to the error vector δ_t and defines the temporal dependency between water storage changes at each time point to previous time steps. This dependency is considered here to be unknown and the matrix Q is estimated along with the unknown state parameters Θ_t within an MCMC procedure that is described in Section ESM- Section 1.2.

Estimation of Q is the main difference between the Bayesian formulation of this study (MCMC-DA) compared to the DMDA method in Mehrnegar et al. (2020). Generally speaking, DMDA also solves the state-space model in a Kalman Filtering (KF, Kalman, 1960) Bayesian approach. To control the temporal dependency in DMDA, a forgetting factor $0 < \lambda \leq 1$ was selected instead of estimating Q as is considered in the implementation in this paper. The forgetting factor was chosen experimentally by defining an effective window size of temporal dependency. The constant was also considered to be the same for different water storage components and for all the grid points within the study area, which is not physically justified as the hydrological characteristics change across the CONUS.

ESM- Section 1.2. MCMC with Gibbs sampling algorithm

MCMC is a sampling method, which follows a Bayesian inference to estimate the full posterior distribution of the parameters of interest, given a set of observations. MCMC sampling provides a class of algorithms for systematic random sampling from high-dimensional probability distributions, where each new sample depends on the existing samples, known as Markov Chain. This property allows the algorithms to focus on the quantity that is being approximated from the distribution, even with a large number of random variables (Van Ravenzwaaij et al., 2018).

The Gibbs sampling (Gelfand and Smith, 1990; Smith and Roberts, 1993) is one of the most common MCMC algorithms, which repeatedly generates a Markov chain of samples from the distribution of each variable, in turn, conditional on the current values of other variables and the data. Gibbs sampling is applicable when the joint distribution of the parameter of interest is difficult to sample from directly, but the conditional distribution of each variable is known and is easier to sample from, which is the case here.

In the linear state-space model (Eqs. (ESM.1) and (ESM.2)), the conditional distribution of the parameter of interest, i.e., Θ_t and Q , is defined by the observation equation (Eq. (ESM.1)) and the state equation

(Eq. (ESM.2)). The Gibbs sampling is selected here to generate samples from the distribution of (i) time varying coefficients ($\Theta_{1:T}$) (ii) the error covariance matrix of $\delta_{1:T}$, i.e., (Q), conditional on the observed data ($Y_{1:T}$) and its error covariance matrix ($V_{1:T}$), and the rest of the unknown parameters, i.e., Q in (i) and $\Theta_{1:T}$ in (ii).

The state-space model given by Eqs. (ESM.1) and (ESM.2) is linear and we assumed that the distribution of GRACE TWSC (observations, Y_t) and hydrological model outputs (Z_t) are Gaussian and independent from each others (though their covariances are assumed to be spatially correlated). Therefore, the conditional posterior distribution of $\Theta_{1:T}$ is a product of the Gaussian Probability Density Functions (PDFs), which can be generated using a standard simulation smoother introduced by [Carter and Kohn \(1994\)](#). Samples generated from the conditional posterior of Q are the product of independent Inverse-Wishart distributions, which are defined on symmetric and positive definite matrices and used generally as the conjugate prior for the covariance matrix of a multivariate normal distribution in the Bayesian inference. In Bayesian theory, the prior is called a conjugate prior for the likelihood function when the posterior distributions of the parameter of interest have the same functional form as the prior. In Bayesian inference conjugate priors are useful because they reduce Bayesian updating to modifying the parameters of the prior distribution, rather than computing integrals, and ensure that the posterior is tractable even after multiplying the likelihood times the prior. Moreover, conjugate prior allows for efficient inference algorithms because the posterior and prior share the same functional form ([Gutiérrez-Pena and Smith, 1995](#)). The implemented Gibbs sampling for the linear state-space model in Eqs. (ESM.1) and (ESM.2) is summarized in what follows.

Step1:

Gibbs sampling is based on the Bayes rule, which uses the knowledge derived from observations (through the likelihood function), to update prior knowledge about the parameters of interest to become posterior belief about them. Therefore, the first step of Gibbs sampling is to define initial states, or prior values, for the unknown state parameters Θ_t , where $t = 0$, and the error covariance matrix of additive innovation δ_t , i.e., $Q^{(i)}$, where i denotes the number of iteration in the Gibbs sampling (it is zero here). Details of prior values for the unknown parameters are explained in [ESM- Section 1.2.1](#).

Step2:

Sample $\Theta_{1:T}(i)$ from the posterior PDF of $\Theta_{1:T}(i)$ conditional on the observed data ($Y_{1:T}$) and its error covariance matrix ($V_{1:T}$), and the covariance matrix of additive innovations δ_t , i.e., $Q^{(i-1)}$. This sampling is defined by:

$$p(\Theta_{1:T}^{(i)} | Y_{1:T}, V_{1:T}, Q^{(i-1)}) = p(\Theta_T^{(i)} | Y_{1:T}, V_{1:T}, Q^{(i-1)}) \prod_{t=1}^{T-1} p(\Theta_t^{(i)} | \Theta_{t+1}^{(i)}, Y_{1:T}, V_{1:T}, Q^{(i-1)}), \quad (\text{ESM.4})$$

where

$$p(\Theta_t^{(i)} | \Theta_{t+1}^{(i)}, Y_{1:T}, V_{1:T}, Q^{(i-1)}) \sim N(\Theta_{t|t+1}^{(i)}, \Sigma_{t|t+1}^{(i)}). \quad (\text{ESM.5})$$

In Eq. (ESM.4) and for the rest of equations in this section, $p(\circ|\ast)$ is used to denote a generic PDF of (a variable such as $\circ$) conditional on (another variable such as $\ast$), while $N(\cdot)$ indicates a Gaussian PDF, and $\Pi(\cdot)$ is an operator to multiply PDF. Forward-filtering backward-smoothing approach, as in [Kitagawa \(1987\)](#), is used to estimate unknown state parameters $\Theta_{t|t+1}$ and their error covariance matrices $\Sigma_{t|t+1}$. Details and the corresponding equations are provided in [ESM- Section 1.2.2](#).

Step3:

Sample $Q^{(i)}$ from the posterior PDF of $Q^{(i)}$ conditional on the observed data ($Y_{1:T}$) and its error covariance matrix ($V_{1:T}$), and $\Theta_{1:T}^{(i)}$ that are derived from Step 2. This sampling is defined by:

$$p(Q^{(i)} | Y_{1:T}, V_{1:T}, \Theta_{1:T}^{(i)}) \sim IW(\bar{Q}^{(i)}, \bar{\nu}), \quad (\text{ESM.6})$$

where

$$\begin{aligned} \bar{\nu} &= T + \nu, \\ \bar{Q}^{(i)} &= Q^{(0)} + \sum_{t=1}^T (\Theta_t^{(i)} - \Theta_{t-1}^{(i)})(\Theta_t^{(i)} - \Theta_{t-1}^{(i)})'. \end{aligned} \quad (\text{ESM.7})$$

In Eq. (ESM.6), $IW(\cdot)$ denotes an Inverse-Wishart PDF, $\bar{Q}$ is the posterior scale matrix, and ν is an initial value that is chosen as the degree of freedom to define the conjugate prior for Q as the product of independent Inverse-Wishart distribution (see ESM- Section 1.2.1), and $\bar{\nu}$ is the posterior value of the degree of freedom. In all these equations T denotes the total number of time steps t .

Step4:

Return to step 2 and continue the iteration until a breaking criterion is satisfied. In order to determine the number of iterations, that defines the convergence of the sampling algorithm, we use a simple graphical method suggested by [Brooks and Gelman \(1998\)](#); [Sinharay \(2003\)](#). This is done by creating a time-series plot for each of the parameters of interest, i.e., Θ_t in Eq. (ESM.1), and Q in Eq. (ESM.3), to view the path traversed by the chains. From the obtained results (figures not shown), we find out that after 10000 iterations, the Gibbs sampling converged to stationary distributions. However, to increase confidence in the process, we select more iterations, i.e., $N=20000$. The reason for this is that after initializing the sampling algorithm with a priori values for the unknown states parameters Θ_t , and the covariance of additive innovations Q , samples from early iterations may not necessarily be representative of the actual posterior distributions. Therefore, the early $M=500$ iterations are discarded as the ‘burn-in’ period. At the end of the Gibbs sampling, $N - M$ number of generated samples for $\Theta_{1:T}$, derived from the forward-filtering backward-smoothing approach (see Eq. (ESM.17) in ESM- Section 1.2.2) are used to estimate the posterior value of unknown state parameters ($\bar{\Theta}_{1:T}$) as:

$$\bar{\Theta}_t = \frac{1}{N - M} \sum_{i=M+1}^N \Theta_t^{(i)} \quad (\text{ESM.8})$$

with $\bar{\Theta}_t = [\bar{\theta}_{1,t}, \bar{\theta}_{1,t}, \dots, \bar{\theta}_{K,t}]$, which are used to update TWSC derived from model ($\hat{Y}_t$) within the observation equation (Eq. (ESM.1)) as:

$$\hat{Y}_t = Z_t \bar{\Theta}_t \quad \text{for } t=1,2,\dots,T. \quad (\text{ESM.9})$$

The Bayesian updated values of W3RA individual water storage estimates $\hat{z}_{k,t,p}$ for $k=1,2,\dots,5$ (corresponding to surface water storage, surface soil water storage (top layer), shallow rooted and deep rooted soil water storage, and groundwater storage), and for each spatial grid point $p = 1, 2, \dots, P$ at each time step $t = 1, 2, \dots, T$ are obtained using $\bar{\theta}_{k,t,p}$ as:

$$\hat{z}_{k,t,p} = z_{k,t,p} \bar{\theta}_{k,t,p} \quad (\text{ESM.10})$$

The values are estimated for 71212 grid points across the entire CONUS, and for $t=1,2,\dots,168$ months covering 2003-2017. In this study, we refer to these estimates as MCMC-DA results. Considering this scale, the application of MCMC-DA within CONUS is computationally expensive. Especially, convergence of the Gibbs sampling might take around 1 month using a general-purpose computer (e.g., 8-core CPU (2.9GHz) with 64GB RAM memory). Therefore, a paralleled implementation of the Gibbs sampling on a High Performance Computing (HPC) system is desirable to maximize the performance of MCMC-DA in large-scale applications.

Figure ESM.1 summarises the work-flow of the MCMC-DA approach to update water storage components of the hydrological model using GRACE TWSC estimates.

The uncertainties of the MCMC-DA water storage components are estimated using the variance of generated samples $\Theta_{1:T}$, i.e., $\Sigma_{1:T}$, which is derived from the forward-filtering backward-smoothing approach in ESM- Section 1.2.2. To this aim, the posterior value of $\Sigma_{1:T}$ is obtained as:

$$\bar{\Sigma}_t = \frac{1}{N - M} \sum_{i=M+1}^N \Sigma_t^{(i)}, \quad (\text{ESM.11})$$

where the diagonal elements of the error covariance matrix of $\bar{\Sigma}_t$, containing $\delta \bar{\theta}_{k,t}^2$ that corresponding to the variance of $\bar{\theta}_{K,t}$, are used to estimate uncertainties of the MCMC-DA water storage estimates. This is done through an error propagation procedure while using the uncertainty of W3RA’s individual water storage

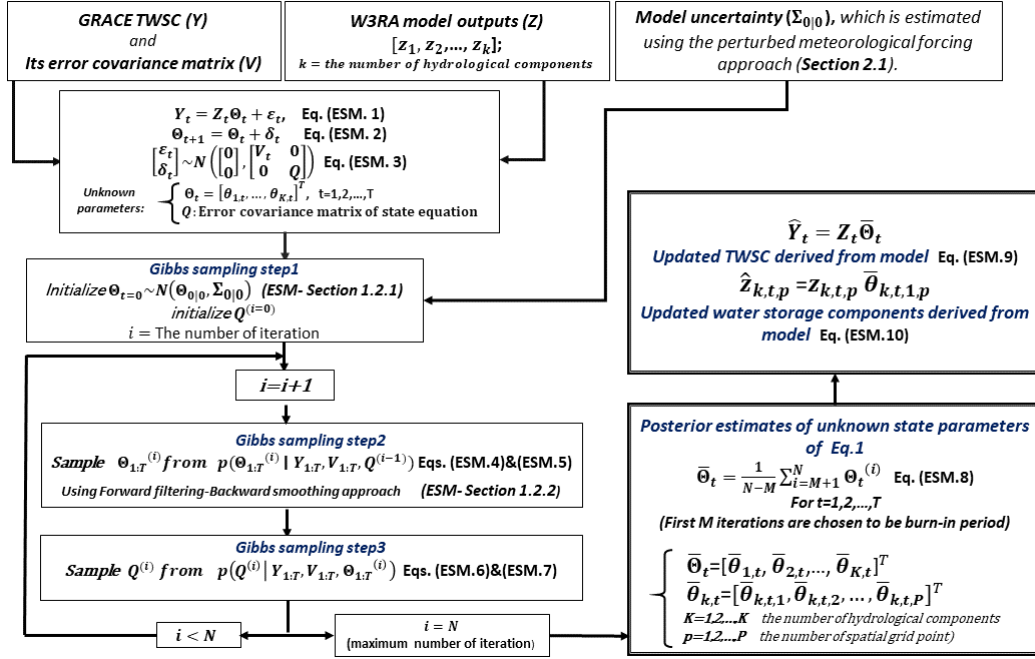


Figure ESM.1: Flowchart of the MCMC-DA method. The framework can accept an arbitrary number of models and it can be extended to accept various type of observations.

estimates ($\delta z_{k,t}^2$) as:

$$\delta \hat{z}_{k,t}^2 = \delta \bar{\theta}_{k,t}^2 \cdot z_{k,t}^2 + \delta z_{k,t}^2 \cdot \bar{\theta}_{k,t}^2, \quad \text{for } k=1,2,\dots,K \quad \text{and} \quad t=1,2,\dots,T, \quad (\text{ESM.12})$$

where $\delta \hat{z}_{k,t}^2$ are the uncertainty of the MCMC-DA water storage estimates for $t=1,2,\dots,T$ and $k=1,2,\dots,5$ corresponded to the $\hat{z}_{k,t}$ derived from MCMC-DA using Eq. (ESM.10).

ESM- Section 1.2.1. Prior values for unknown parameter in Gibbs sampling

The first step of Gibbs sampling to solve the state-space model, given by Eq.s (ESM.1) and (ESM.2), is to define initial states, or prior values, for the unknown state parameters Θ_t , where $t = 0$, and the error covariance matrix of additive innovation δ_t , i.e., $Q^{(i)}$, where i denotes the number of iteration in the Gibbs sampling (it is zero here). As it was discussed in Section ESM- Section 1.2, since the state-space equation is linear, and with assuming the distribution of GRACE TWSC (observations, Y_t) and hydrological model outputs (Z_t) to be Gaussian and independent from each others, the conditional posterior distribution of $\Theta_{1:T}$ in Eq. (ESM.4) is a product of their (Gaussian) Probability Density Functions (PDFs). Therefore, the prior value of Θ_0 is Gaussian distributed and is defined as:

$$\Theta_0 \sim N(\Theta_{0|0}, \Sigma_{0|0}), \quad (\text{ESM.13})$$

where $N(\cdot)$ represents a Gaussian (Normal) distribution, and $\Theta_{0|0}$ and $\Sigma_{0|0}$ are the mean and variance of Θ_0 . In this study, $\Theta_{0|0}$ is chosen to be 1, because in theory, the summation of individual water storage values Z_t must be equal to the GRACE TWSC observation Y_t in Eq. (ESM.1). Uncertainty of simulated water storage simulations of the original W3RA (see Section 2.1) are stored in $\Sigma_{0|0}$, which are needed for computing updates by the Gibbs sampling in Step 2.

In Eq. (ESM.2), Q is the error covariance matrix of additive innovation δ_t , and defines the temporal dependency between water storage states Θ_t at each time point to the previous time steps. Considering Eq. (ESM.13), the conjugate prior for Q can be estimated by an independent Inverse-Wishart distribution

($IW(\cdot)$) as:

$$Q^{(0)} \sim IW(S, \nu), \quad (\text{ESM.14})$$

where S is the scale matrix, and ν is the degree of freedom [Schuurman et al. \(2016\)](#), which is set to 1 that is the sample rate of monthly data (for more information please see [Primiceri, 2005](#)). Following [Cogley \(2005\)](#); [Primiceri \(2005\)](#); [Schuurman et al. \(2016\)](#), the scale matrix S is chosen to be a constant fraction of the variance of the initial values Θ_0 as $K_Q^2 \Sigma_{0|0}$. Therefore, Eq. (ESM.14) is rewritten as:

$$Q^{(0)} \sim IW(K_Q^2 \Sigma_{0|0}, 1). \quad (\text{ESM.15})$$

In Eq. (ESM.15), $\Sigma_{0|0}$ is the covariance matrix that is derived from the ensemble of W3RA (see Eq. (ESM.13)). Following [Primiceri \(2005\)](#), we chose $K_Q = 0.01$, which allows that Θ_t to be temporally variable. More details about selecting different prior values and their impact on the Inverse-Wishart estimation can be found in [Schuurman et al. \(2016\)](#). Prior values chosen for Θ_0 and $Q^{(0)}$ are then used in Step 2 and Step 3 of Gibbs sampling in Eqs. (ESM.5) and (ESM.7), respectively.

ESM- Section 1.2.2. The forward-filtering backward-smoothing approach

This appendix describes the forward-filtering backward-smoothing approach, which is used in the second step of the Gibbs sampling (see Eq. ESM.4) to generate samples of $\Theta_{1:T}$ (unknown state parameters in the state-space model given by Eqs. (ESM.1) and (ESM.2)) from the PDF of $\Theta_{1:T}$ conditional on the observed data ($Y_{1:T}$) and its error covariance matrix ($V_{1:T}$), and the rest of unknown parameters, i.e., Q .

In the rest of this section, $\Theta_{t|t}$ and its variance $\Sigma_{t|t}$, as well as $\Theta_{t|t-1}$ and $\Sigma_{t|t-1}$ are defined as:

$$\begin{aligned} \Theta_{t|t} &= E(\Theta_t | Y_{1:t}, V_{1:t}, Q), \\ \Sigma_{t|t} &= \text{Var}(\Theta_t | Y_{1:t}, V_{1:t}, Q), \\ \Theta_{t|t-1} &= E(\Theta_t | Y_{1:t-1}, V_{1:t-1}, Q), \\ \Sigma_{t|t-1} &= \text{Var}(\Theta_t | Y_{1:t-1}, V_{1:t-1}, Q), \end{aligned} \quad (\text{ESM.16})$$

where t represents the monthly time steps between 1 and T , and $E(\circ|*)$ and $\text{Var}(\circ|*)$ denote the mean value and the variance for the normal distribution of $(\circ)$ conditional on $(*)$.

The forward-filtering backward-smoothing is a recursive approach that consists of two steps:

(1) **Forward-filtering:** where we use a standard Kalman-Filter (KF) following [Carter and Kohn \(1994\)](#) to recursively estimate $\Theta_{t|t}$ and $\Sigma_{t|t}$, for $t=1,2,\dots,T$, given the prior values of $\Theta_{t|t}$ and $\Sigma_{t|t}$ when $t=0$ ($\Theta_{0|0}$ and $\Sigma_{0|0}$, see ESM- Section 1.2.1). The KF recursion is presented in the following equations as:

$$\begin{aligned} \Theta_{t|t-1} &= \Theta_{t-1|t-1}, \\ \Sigma_{t|t-1} &= \Sigma_{t-1|t-1} + Q, \\ K_t &= \Sigma_{t|t-1} Z_t' (Z_t \Sigma_{t|t-1} Z_t' + V_t)^{-1}, \\ \Theta_{t|t} &= \Theta_{t|t-1} + K_t (y_t - Z_t \Theta_{t|t-1}), \\ \Sigma_{t|t} &= \Sigma_{t|t-1} - K_t Z_t \Sigma_{t|t-1}. \end{aligned} \quad (\text{ESM.17})$$

The last elements of the KF recursion when $t=T$, i.e., $\Theta_{T|T}$ and $\Sigma_{T|T}$, are used to generate the samples of Θ_T in Eq. (ESM.18) as:

$$\Theta_t \sim N(\Theta_{t|t}, \Sigma_{t|t}) \quad (\text{ESM.18})$$

where $N(\cdot)$ denotes the normal distribution of Θ_t with the mean value of $\Theta_{t|t}$ and the variance of $\Sigma_{t|t}$.

(2) **Backward-smoothing:** The outputs of step (1) (i.e., $\Theta_{t|t-1}$, $\Sigma_{t|t-1}$, $\Theta_{t|t}$, and $\Sigma_{t|t}$ for $t=1,2,\dots,T$), and the generated sample of Θ_T , derived from Eq. (ESM.18) are then used in Eq. (ESM.19) to update $\Theta_{t-1|t}$ and $\Sigma_{t-1|t}$, and generate samples of Θ_{t-1} , for $t = T, T-2, \dots, 1$. For a generic time t , the updating formulas of the backward recursion smoother can be written as:

$$\begin{aligned}
\Theta_{t-1|t} &= \Theta_{t-1|t-1} + \Sigma_{t-1|t-1} \Sigma_{t|t-1}^{-1} (\Theta_t - \Theta_{t-1|t-1}), \\
\Sigma_{t-1|t} &= \Sigma_{t-1|t-1} - \Sigma_{t-1|t-1} \Sigma_{t|t-1}^{-1} \Sigma_{t-1|t-1}, \\
\Theta_{t-1} &\sim N(\Theta_{t-1|t}, \Sigma_{t-1|t}).
\end{aligned} \tag{ESM.19}$$

The backward recursion smoother is started from time T and continues until time 1. The output of the Forward-filtering Backward-smoother approach is the generated samples of $\Theta_{1:T}$, conditional on the observed data ($Y_{1:T}$) and its error covariance matrix ($V_{1:T}$), and the unknown covariance matrix of the additive innovation Q , which is defined as $p(\Theta_t^{(i)} | \Theta_{t+1}^{(i)}, Y_{1:T}, V_{1:T}, Q^{(i-1)})$ in Eq. (ESM.5) in ESM- Section 1.2. The generated samples of $\Theta_{1:T}$ are then used in the third step of Gibbs sampling (see ESM- Section 1.2) to generate samples of the unknown covariance matrix of additive innovations, i.e., shown by Q in Eq. (ESM.6).

ESM- Section 2. Comparison between GRACE and W3RA TWSC over CONUS

In this section, we provide an overview of the comparison of variance between GRACE TWSC (used as observations) and W3RA (whose individual water storage estimates are used as a priori information in the MCMC-DA approach). Grid based linear trends and annual amplitudes of TWSC correspond to $\sim 36\%$ and $\sim 53\%$ of the total variance over the CONUS, respectively (see Fig. ESM.2). These two components are also shown for the estimated individual water storage changes of the original W3RA model and those updated by the MCMC-DA approach. These results are presented in the following sections.

In order to compute the linear trend along with annual and semi-annual cycles, a regression equation as $l(t) = a + b t + c \cos(2\pi t) + d \sin(2\pi t) + e \cos(4\pi t) + f \sin(4\pi t) + \epsilon(t)$ is used. Here, $l(t)$ contains TWSC time series in each grid (as in Fig. ESM.2) or individual water storage time series, and t is a time vector that varies between 2003 and 2017. To estimate the unknown coefficients (shown by $\hat{a}$, $\hat{b}$, $\hat{c}$, $\hat{d}$, $\hat{e}$, and $\hat{f}$), a least squares approach is used while assuming $\epsilon(t)$ to be the vector of residuals. We consider $\hat{b}$ as the linear trend (in mm/yr) and $\sqrt{\hat{c}^2 + \hat{d}^2}$ as amplitude of the annual cycle (in mm). The non-parametric Wilcoxon-Mann-Whitney statistical test (WMW, *Fay and Proschan, 2010*) with 95% and 99% significant levels is used to evaluate the statistical significance of the estimated unknowns.

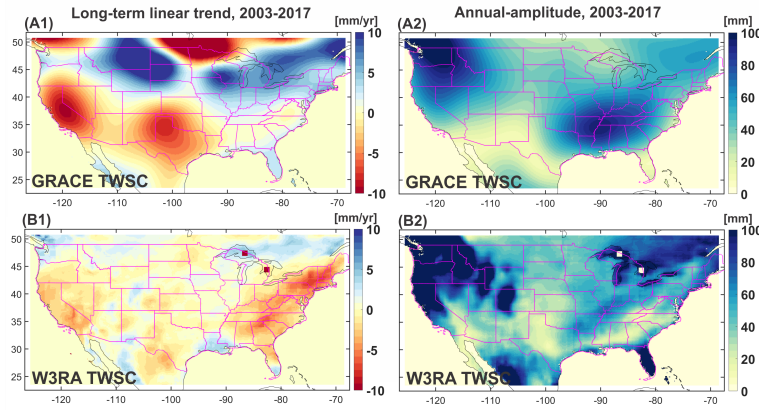


Figure ESM.2: Linear trend [mm/yr] and annual-amplitude [mm] fitted to the GRACE TWSC and the original W3RA TWSC within the CONUS, covering the period of 2003-2017. The plots in (A1) and (A2) correspond to GRACE TWSC, while (B1) and (B2) correspond to W3RA TWSC.

TWSC are shown in Fig. ESM.2, where considerable differences are found between the original W3RA TWSC and GRACE TWSC in terms of both linear trend (compare Fig. ESM.2 A1 and Fig. ESM.2 B1) and annual-amplitude (compare Fig. ESM.2 A2 and Fig. ESM.2 B2) within different parts of the CONUS,

where large differences in the linear trend are found in the west and south, e.g., California and Texas that are influenced by irrigation and drought (*Famiglietti et al., 2011; Scanlon et al., 2012; Rodell et al., 2018*). GRACE indicates larger trends in these regions with the mean value of -9.47 ± 1.02 mm/yr in California and -7.65 ± 1.32 mm/yr in Texas, relative to those of W3RA with the mean value of -3.15 ± 0.86 mm/yr and -3.86 ± 1.24 mm/yr within California and Texas, respectively. Within the northern CONUS (e.g., Montana, North Dakota, and South Dakota), similar results are found with higher positive trends from GRACE ($\sim 9.48 \pm 1.78$ mm/yr) than those of W3RA ($\sim 2.61 \pm 0.86$ mm/yr). More pronounced changes in GRACE TWSC within the northern CONUS are attributed to high-intensity rainfall and flooding events (e.g., the 2011's flood in the Missouri River, *Zhang and Schilling, 2006; Zheng et al., 2014; Reager et al., 2014*).

Large differences in the annual-amplitude of TWSC are found within the Great Lakes and in the south-eastern CONUS, with the mean values of 76.23 ± 4.08 mm derived from GRACE TWSC and 34.65 ± 3.21 mm from W3RA TWS. In contrast, the annual-amplitude of W3RA TWSC within Florida is estimated up to 96.5 ± 5.34 mm, which is much greater than that of estimated by GRACE TWSC (20 ± 3.5 mm). The differences in the seasonality of TWSC, between modeled and measured, can be related to errors in the forcing data and uncertainty in model parameters to control these values (*Van Dijk, 2010; Van Dijk et al., 2011*), which is the case for most available models as demonstrated by *Scanlon et al. (2018)* and *Mehrnegar et al. (2020)*. The comparison in Fig. ESM.2 indicates that GRACE TWSC has the potential to be used for modifying the W3RA's water storage simulations in terms of long-term trends and seasonality. We will also show that the modification is even beyond these components and some climate modes such as that of ENSO is influenced after implementing MCMC-DA.

ESM- Section 3. Supplementary Figures

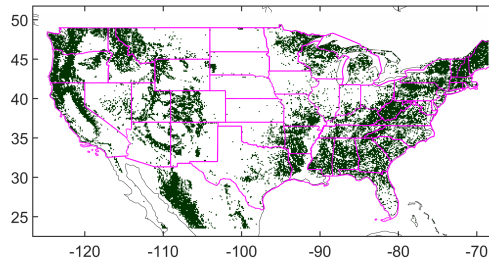


Figure ESM.3: The forest land distribution with 80% density across the CONUS, which is provided using the ESA CCI's Land Cover (LC) data (v2.0.7) <http://maps.elie.ucl.ac.be/CCI/viewer/download.php>. The original LC data is a consistent global LC map at 300 m spatial resolution, on an annual basis from 1992 to 2015. For this study, we extracted the 2015's LC data within the CONUS, with the spatial resolution of 12.5 km.

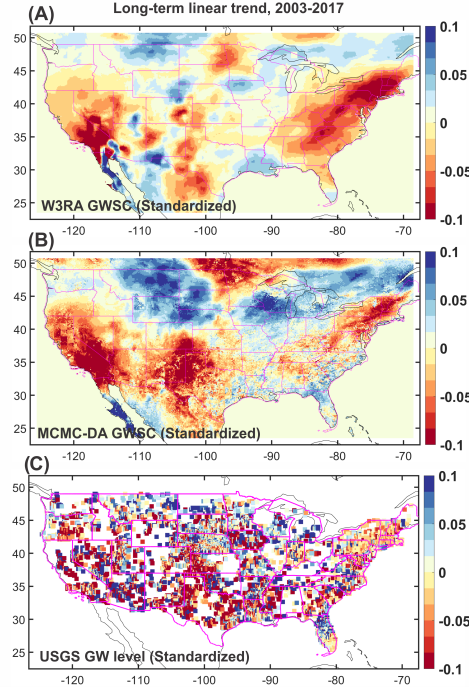


Figure ESM.4: Linear trend fitted to the standardized (A) groundwater storage estimates from the original W3RA, (B) MCMC-DA groundwater storage, and (C) USGS groundwater level observations covering the period of 2003-2017.

ESM- Section 4. Correlation coefficients of groundwater storage changes with USGS observation

In order to show the impact of integrating GRACE data into modeled groundwater estimates beyond the linear trend and seasonal components, correlation coefficients are estimated between standardized USGS groundwater level time series and those of the original W3RA and MCMC-DA. Computations are done after removing the trend and annual amplitudes and the results are shown in Fig. ESM.5 A (USGS versus W3RA) and Fig. ESM.5 B (USGS versus MCMC-DA), where improvements are found in 76% of the grid points. The greatest improvements are located within the southeastern CONUS, including Florida and Louisiana, and within the central CONUS, including Texas, Utah, and Nebraska (i.e., more than 40% improvements on average in both central and southeastern CONUS).

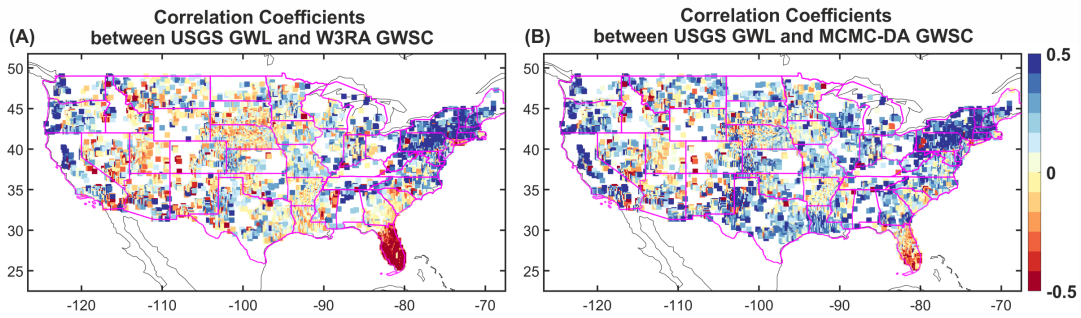


Figure ESM.5: Correlation coefficients (A) between the USGS groundwater level observations and groundwater storage estimates from the original W3RA; and (B) between the USGS groundwater level measurements and the groundwater storage estimates of MCMC-DA. The results correspond to the period of 2003-2017.

ESM- Section 5. Correlation coefficients and RMSD of soil water storage changes with the ESA CCI soil moisture products

Temporal correlation coefficients and the Root Mean Squares of Differences (RMSD) between ESA CCI soil moisture product and the top-layer soil water changes derived from W3RA and MCMC-DA are shown in Fig. ESM.6, where the results indicate average improvements of 67% and 73%, respectively. Positive correlation coefficients (higher than 0.6) are found within $\sim 90\%$ of the CONUS between the ESA CCI and both W3RA and MCMC-DA estimates. WMW statistical test (at 95% confident level) indicates that except the southeastern part of the CONUS (i.e., Alabama, Georgia, South Carolina, Mississippi, and Louisiana states) the correlations values are increased after implementing MCMC-DA. A comparison between the forest coverage across the CONUS (Fig. ESM.3) and Figs. ESM.6 A1 and A2 indicates that the minimum correlation coefficients (less than 0.4) is found within the forest regions, where both W3RA and ESA-CCI are relatively weak to represent the true signals. This is also reflected in the RMSD plots, where except the southeastern CONUS the differences are decreased from ~ 30 mm to ~ 5 mm in the central and western parts of CONUS (see Figs. ESM.6 B1 and B2).

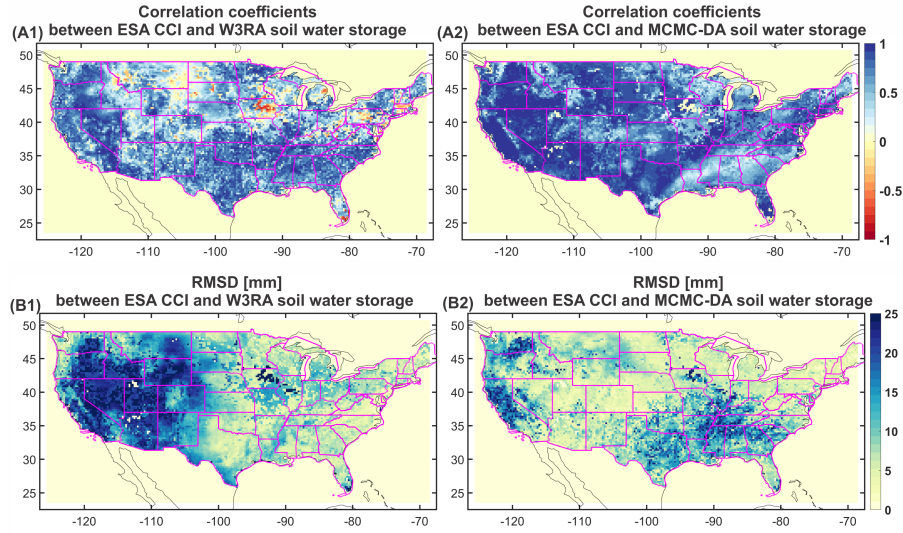


Figure ESM.6: Correlation coefficients and RMSD between the ESA CCI soil moisture products and in (A1) and (B1), soil water storage estimates from the original W3RA; and in (A2) and (B2), the MCMC-DA's soil water storage. The computations use the whole period of 2003-2017.

ESM- Section 6. Independent Component Analysis (ICA) to extract the ENSO modes from water storage changes

ICA is a higher (than second) order statistical decomposition technique, which incorporates statistical moments of available data samples in its computational procedure to derive empirical components that are statistically as independent as possible. The ICA technique that is applied here follows the formulation of [Forootan and Kusche \(2012, 2013\)](#), in which the ICA is a rotated extension of the Principal Component Analysis (PCA, [Forootan, 2014](#), chapter 3). For this, an orthogonal rotation matrix $\mathbf{R}$ is computed in an optimization procedure, which can be used to rotate the PCA's orthogonal components make them as statistically as independent as possible. The ICA decomposition of the data matrix $\mathbf{X}$ can be formulated as:

$$\mathbf{X} = \mathbf{PRR}^T\bar{\mathbf{E}}^T, \quad (\text{ESM.20})$$

where $\mathbf{PR}$ and $\mathbf{ER}$ are rotated components and always only one of them is independent. For the temporal ICA technique, columns of $\mathbf{PR}$, and for the Spatial ICA method, columns of $\mathbf{ER}$ are statistically as independent as possible (details of the Temporal and Spatial ICA techniques and the approach to estimate $\mathbf{R}$ can be found in [Forootan, 2014](#), chapter 4).

Previous experiments (e.g., [Forootan et al., 2016, 2018](#)) indicate that the Temporal ICA technique can better separate the ENSO modes from other components, e.g., trends and cyclic signals. In this paper, the Temporal ICA is applied to extract the ENSO modes but its name is simply called ICA. For this, we first reduce the linear trend and seasonal cycles from the time series of water storage and precipitation estimates across the CONUS. The residuals are then replaced as $\mathbf{X}$ in Eq. [ESM.20](#). Our results indicate that the first two ICA modes are appeared to be well correlated with the Niña 3.4 ENSO index. Therefore, these modes are shown and interpreted in [Section 4.3](#), i.e., Figs. [5](#), [6](#), and [7](#).

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